COMS BC1016
Introduction to Computational Thinking and Data Science

Lecture 12: Hypothesis Testing

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October 19, 2026

The Next Few Weeks

Today

10/19	12 - Hypothesis Testing			
10/21	*Midterm Review* (TA Review Session)	*No Lab*		
10/26	**Midterm**			
10/28	Special Topics: Data Privacy (Guest Lecture by Madiha Zahrah Choksi)	*No Lab*		HW 4 Due
11/2	*Academic Holiday*			
11/4	13 - Statistical Signifi nce	Lab 6 - Examining the Therapeutic Touch		
11/6	-			**Final Project Group Declaration** Due

No labs!

Say who you want to work with (or if you want me to find you a group)

Possible extra credit opportunity

Midterm Reminders

- Midterm is next week on **Monday, October 26**
  - Exam will cover material up until last week Wednesday
  - TAs will lead a midterm review next class (October 21)
- In-class paper exam
  - You can bring a single 5”x8” index card with notes (to be submitted along with your exam)
    - Front and back, can be typed or handwritten
    - You don’t need to use the index cards I handed out but you can

Upcoming Office Hours

- Monday, Oct 19:
  - 3pm-4pm Yeonji Baek (Milstein 502)
  - 4pm-5pm Eysa Lee (Milstein 512)
- Tuesday, Oct 20:
  - 9am-10am Minh Le (Milstein 502)
- Wednesday, Oct 21:
  - 2:30pm-3:30pm Aditya Pendyala (Milstein 502)
  - 4pm-5pm Aarya Dharm (Milstein 502)
- Thursday, Oct 22:
  - 2:30pm-3:30pm Eysa Lee (Zoom)

Last Time: Models

Models

A **model** is a set of assumptions about data

- In data science, many models involve assumptions about the processes that involve randomness

- **Key question**: Does the model fit the data?

Approach to Assessment

- If we can simulate the data according to the assumptions of the model, we can learn what the model predicts

- We can then compare the predictions to the data that we observed

- If the data and the model's predictions are not consistent, that is evidence against the model

Steps in Assessing a Model

1. **Choose a statistic** to measure the discrepancy between the model and data

2. **Simulate the statistic** under the assumptions of the model

3. **Compare** the data to the model's predictions

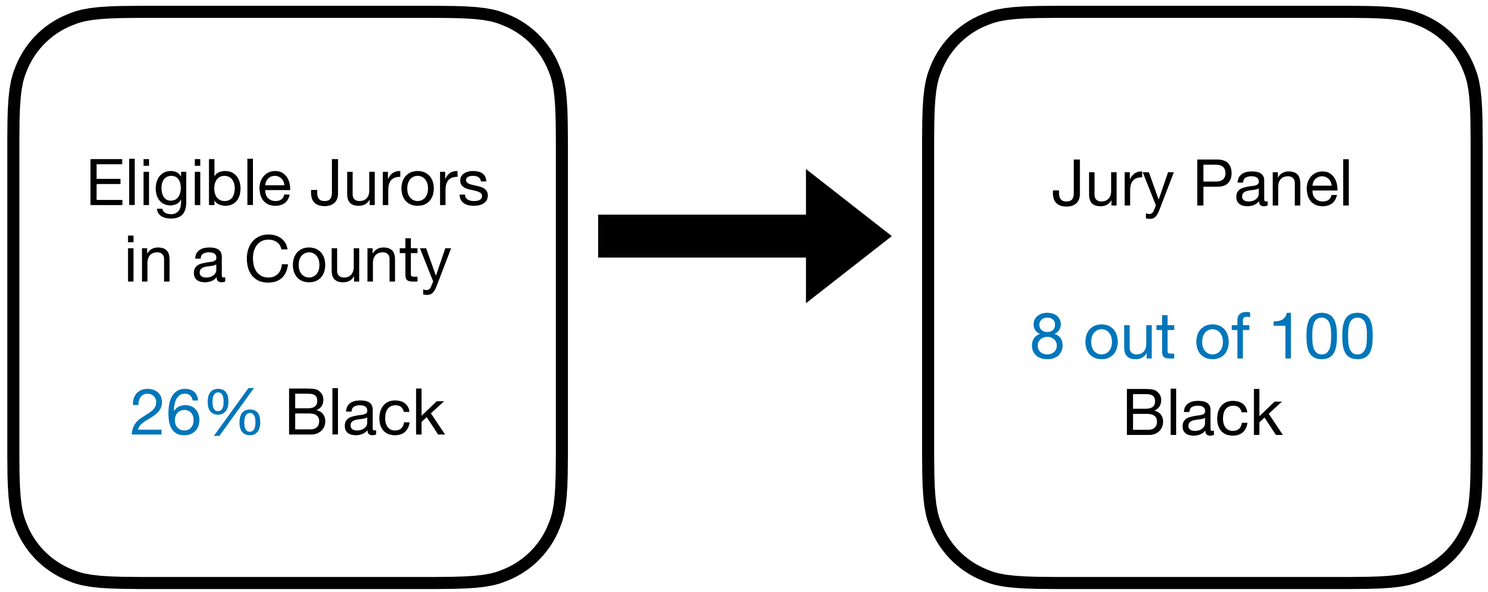
   - Draw a histogram of simulated values of the statistic

   - Compute the observed statistic from the real sample

4. If the observed statistic is far from the histogram, that is evidence against the model

Swain v Alabama

Question: Would an 8% black jury be a realistic outcome if jury section were truly unbiased?



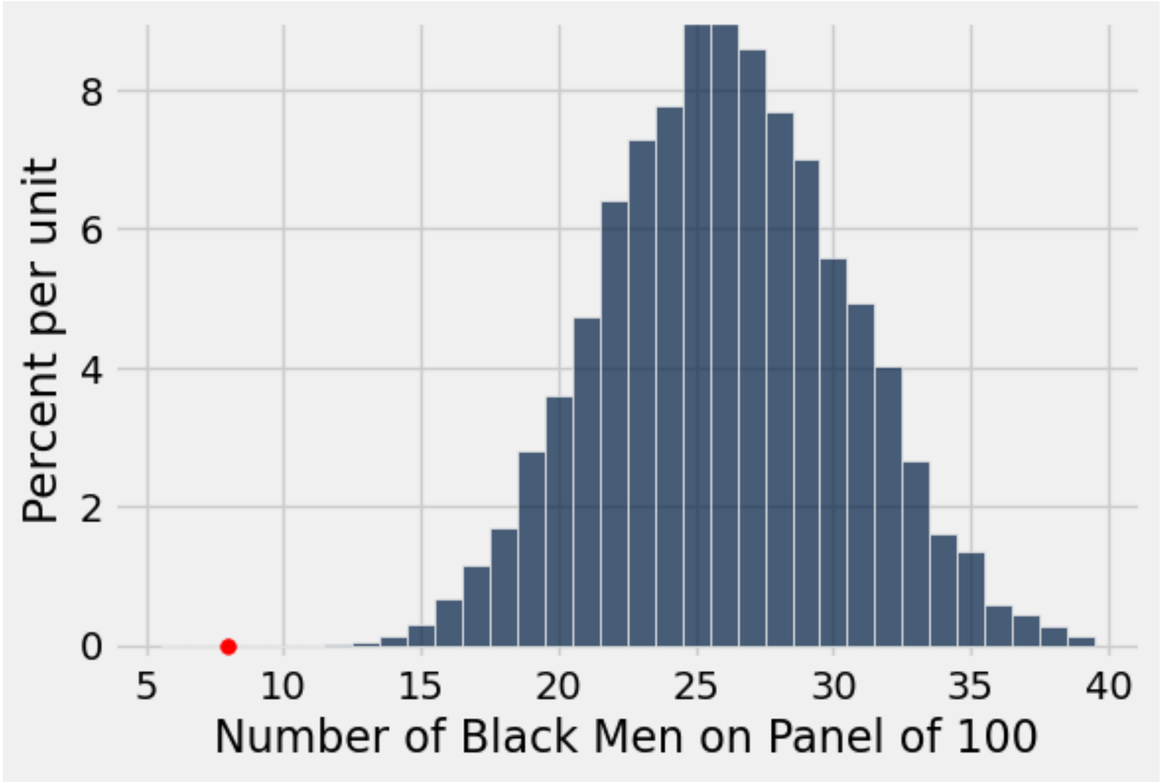
What statistic did we use to measure the discrepancy between the model and the data?

Swain v Alabama Example

Our statistic was “the number of Black panelists in a random sample of 100 people from the population”

Used `sample_proportions` to simulate sampling 10,000 jury panels

Comparing the observed data to the model’s predictions, 8% Black panelists was **highly unlikely** under random sampling



More on Models

Gregor Mendel

- Austrian monk interested in botany
- In the 1850s, experimented with pea plants
  - Self-pollinated plants and collected seeds until plants had similar characteristics each generation (e.g., flower color, plant height)
  - He crossed plants with different characteristics and had the offspring self-pollinate



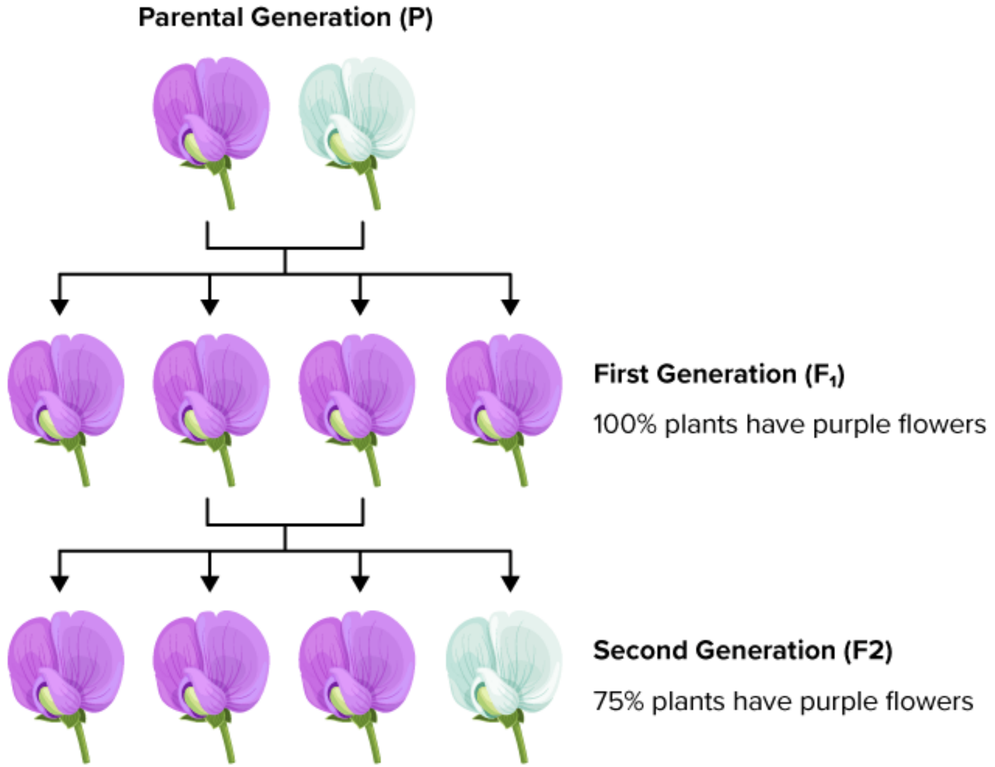
Photo By Unknown author - http://0.tqn.com/d/biology/1/0/l/e/3244238.jpg, Public Domain, https://commons.wikimedia.org/w/index.php?curid=33347279

Mendel's Genetic Model

- Pea plants can either have purple flowers or white flowers (but not a mix)
- Mendel theorized a pattern in the frequency of colors
- Mendel grew 929 crossed plants, and observed 709 had purple flowers

Mendel's Model: Crossed plants have a 75% chance of having purple flowers
regardless of the color of other plants

Question: Is this model accurate?



Steps in Assessing a Model

1. **Choose a statistic** to measure the discrepancy between the model and data

2. **Simulate the statistic** under the assumptions of the model

3. **Compare** the data to the model's predictions

   - Draw a histogram of simulated values of the statistic

   - Compute the observed statistic from the real sample

4. If the observed statistic is far from the histogram, that is evidence against the model

Choosing a Statistic

- Take a sample and see what percentage are purple flowering
- If that percentage is much larger or much smaller than 75, then that is evidence against the model
  - Notice this is another way of measuring the **distance** from 75
- Statistic:

$$\text{distance} = \left|\text{sample percentage of purple-flower plants} - 75\right|$$

- If the distance is large, that is evidence against the model

Demo: Mendel’s Model

Two Viewpoints

- Swain v Alabama
  - **Model**: The people on the jury panels were selected at random from the eligible population
  - **Alternative viewpoint**: No, jury selection was biased against Black men
- Mendel's Flowers
  - **Model**: Each plant has a 75% change of having purple flowers
  - **Alternative viewpoint**: No, it doesn't

Practice: Choosing a Statistic

In each of (a) and (b), choose a statistic that will help you decide between the two viewpoints.

Data: the results of 400 tosses of a coin

(a) "This coin is fair" vs "No, it's not"

(b) "This coin is fair" vs "No, it's biased towards heads"

Breaking down the question

For both (a) and (b): “The coin is fair”

- The percent of heads in 400 tosses is a good starting point
- What percent of heads suggests “fair”?

For (a): What would suggest not fair?

For (b): What would suggest biased towards heads?

Practice: Choosing a Statistic

(a) Very large or very small values of the percent of heads suggest “not fair”:

- Similar to Mendel’s flowers, we can look at the **distance** between percent of heads and 50%

- Statistic: $\text{distance} = \left|\text{percent of heads} - 50\,\%\right|$

  - Large values suggest “not fair”

(b) Large values of the percent of heads suggest “biased towards heads”

- Statistic: percentage of heads

Comparing Distributions

Jury Selection in Alameda County

Racial and Ethnic Disparities in Alameda County Jury Pools

A Report by the ACLU of Northern California

October 2010

Jury Selection in Alameda County

- In 2010, ACLU of Northern California reported that racial and ethnic groups were not properly represented in jury panels in Alameda County, CA

  - 11 felony trials over 2 years (2009 and 2010)

  - Collected demographic data on the 1453 panelists and compared to eligible jurors in the county

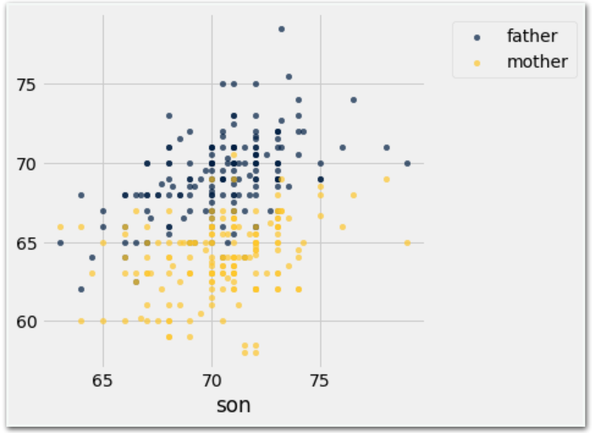
Overlaid Graphs

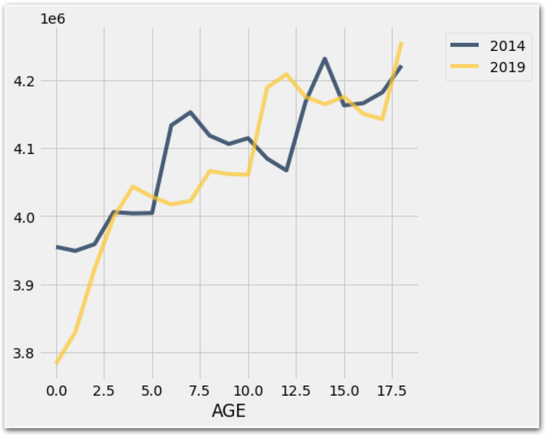
- Drawing two datasets on the same scatter plot, line plot, or bar chart allows visual comparison

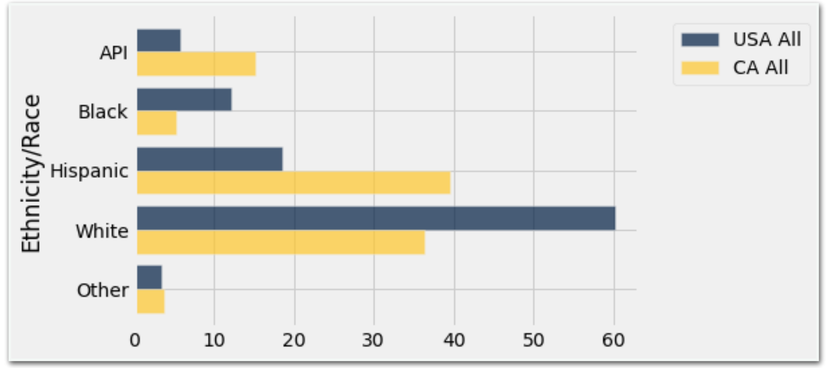
- Overlaid graphs must represent the same variables and be measured in the same units

    - One column must also be a common axis for both datasets

- Can easily be created by specifying the common axis to graph both datasets







DEMO

Distance between Distributions

- In this data, we're looking at a categorical distribution of ethnicities

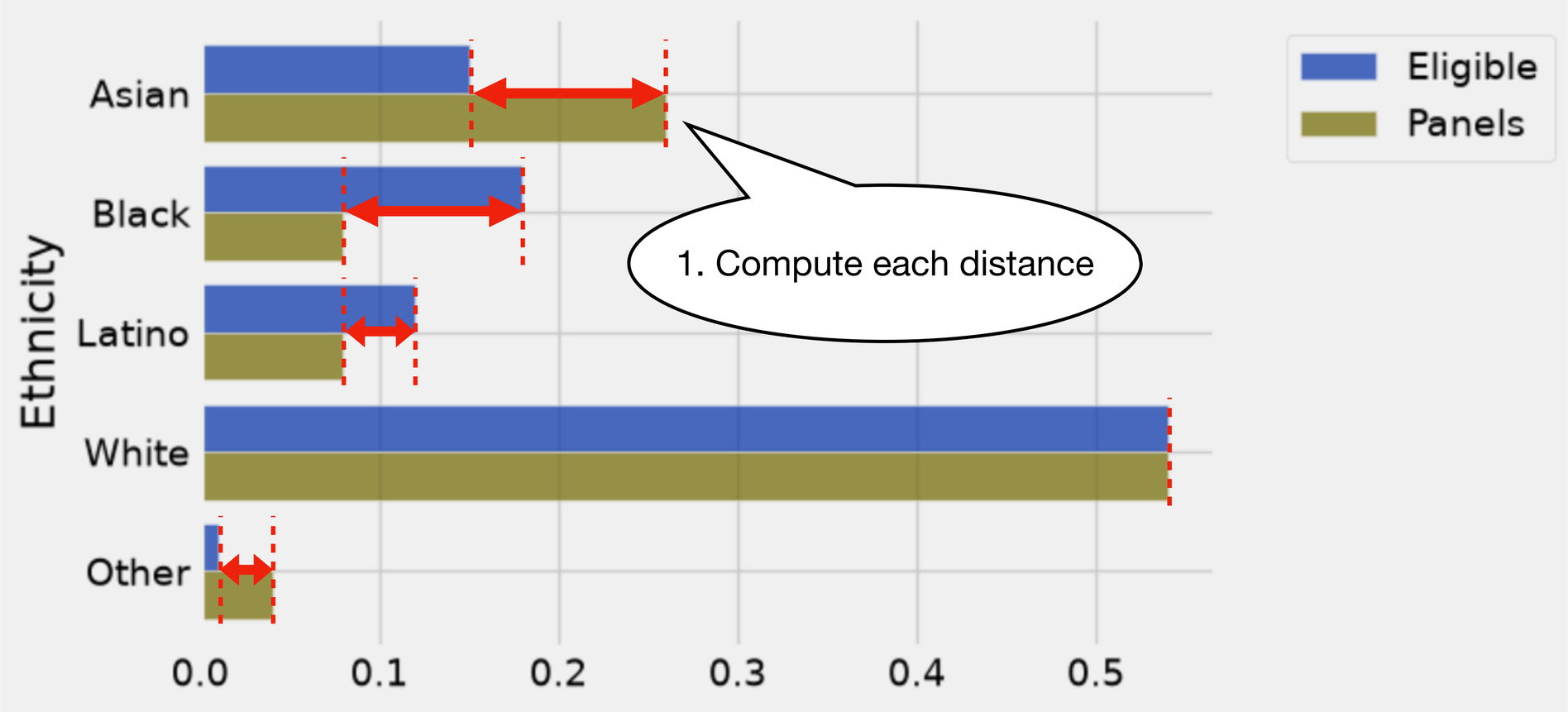
- How can we tell if the distribution of ethnicities of the panels is "close" to that of the eligible jurors?

  - When it's two values, we compute the distance:
    $$\text{distance}(x, y) = |x - y|$$

  - For this, we'll do a generalized version of distance called **Total Variation Distance**

Ethnicity	% in Population
Asian	15
Black	18
Latino	12
White	54
Other	1

Total Variation Distance



2. Compute the absolute value of each distance
3. Sum the absolute values
4. Divide the sum by 2

Summary of the Method

To assess whether a sample was drawn randomly from a known categorical distribution:

- Use TVD as a statistic because it measures the distance between categorical distributions
- Sample at random from the population, and compute the TVD from the random sample. Repeat many times
- Compare:
  - Empirical distribution of simulated TVDs
  - Actual TVD from the sample in the study

Demo: Jury Selection

Hypothesis Testing

Testing Hypotheses

- A test chooses between two views of how data were generated
- The views are called hypotheses
- Hypothesis Testing**:** A statistical test in which we choose between two potential views
  - The test picks the hypothesis that is better suited by the observed data

Null and Alternative

The method only works if we can simulate data under one of the hypotheses

- **Null Hypothesis:** A well defined chance model about how the data was generated.
    - We can simulate data under the assumptions of this model (“under the null hypothesis”)
- **Alternative Hypothesis**: A different view about the origin of the data
    - Doesn’t say how or why the chance model isn’t right, just that it isn’t right

Test Statistic

The statistic we choose to simulate to decide between the two hypotheses

Questions before choosing the statistic:

- What values of the statistic will make us lean towards the null hypothesis?
- What values will make us lean towards the alternative?
    - Preferably the answer should just be “high”
    - Try to avoid “both high and low”

Prediction under the Null Hypothesis

- Simulate the test statistic under the null hypothesis
- Draw the histogram of the simulated values
  - This displays the empirical distribution of the statistic under the null hypothesis
- It is a prediction about the statistic made by the null hypothesis
  - It shows all the likely values of the statistic and how likely they are **(if the null hypothesis is true)**
- Remember, probabilities are approximate because we can't generate all possible random samples

Conclusion of the Test

Resolve choice between null and alternative hypotheses

- Compare the observed test statistic and its empirical distribution under the null hypothesis

- If the observed value is **not consistent** with the distribution, then the test favors the alternative (“data is consistent with the alternative”)

Whether a value is consistent with a distribution:

- A visualization may be sufficient

- If not, there are conventions about “consistency” (we will see soon!)

Next Class

- Today
  - Hypothesis Testing
- Wednesday
  - Midterm Review
- Monday
  - Midterm
- Wednesday
  - Special Topics